

THEORY AND METHODS
VARIANCE AND BIAS DESIGN CRITERIA FOR
ESTIMATING THE TURNING POINT FROM
THE ONE-FACTOR QUADRATIC MODEL

R.L.J. Coetzer¹

Sasol Technology, PO Box 1, Sasolburg 1947, South Africa

e-mail: *roelof.coetzer@sasol.com*

and

L.P. Fatti

University of the Witwatersrand, PO Wits, Johannesburg 2050, South Africa

e-mail: *paulfatti@gmail.com*

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Summary: In this paper we derive and evaluate optimum experimental designs for estimating the turning point, and the corresponding optimal response, from a single factor quadratic model. Since these estimates are nonlinear functions of the parameters, both variance and bias properties are important in their estimation. Therefore, we address the problem of selecting design points for estimating the turning point, and the corresponding optimal response, with minimum variance and bias. New design criteria are presented for deriving locally symmetric and non-symmetric, non-singular and Bayesian optimum designs.

1. Introduction

Let y be a random response variable with constant variance σ^2 whose expected value depends on the level of a controllable factor x :

$$\eta = E(y|x) = f(\underline{x})^T \underline{\tau} \text{ and } V(y|x) = \sigma^2 \quad (1)$$

¹ Corresponding author.

We assume that the relationship is known to be quadratic and convex downwards but is otherwise unknown. That is, we consider the model

$$\eta = \tau_0 + \tau_1 x - \tau_2 x^2 \quad (2)$$

where the values of $\underline{\tau} = (\tau_0, \tau_1, \tau_2)^T$, $\tau_2 > 0$, are unknown and $f(\underline{x})^T = (1, x, -x^2)$. If $\hat{\underline{\tau}} = (\hat{\tau}_0, \hat{\tau}_1, \hat{\tau}_2)^T$ is the least squares estimator of $\underline{\tau} = (\tau_0, \tau_1, \tau_2)^T$, then the optimal turning point, γ , where the response η attains its maximum value is estimated as

$$\hat{\gamma} = \frac{\hat{\tau}_1}{2\hat{\tau}_2} = g_1(\hat{\tau}), \text{ say,} \quad (3)$$

and the corresponding optimal response, y_0 , is estimated as

$$\hat{y}_0 = \hat{\tau}_0 + \frac{\hat{\tau}_1^2}{4\hat{\tau}_2} = g_2(\hat{\tau}), \text{ say,} \quad (4)$$

If we assume that the observation errors are independent normally distributed with common variance σ^2 , (3) and (4) are also the maximum likelihood estimators. The aim is to select N levels of $x : x_1 \leq x_2 \leq \dots \leq x_N$, at least three of which are distinct, for estimating γ and y_0 with minimum variance and bias. Note that, in the one-factor quadratic model, the turning point yields the unique optimum. Without loss of generality, we consider $x_u \in [-1, 1]$, $u = 1, 2, \dots, N$; if necessary, this range can be achieved by scaling. σ^2 is taken as unity since it has no relevance to the design structure.

Optimal experimental designs and the criteria to derive them have received considerable attention in the past (Silvey, 1980; Atkinson, 1988; Atkinson et al., 2007). Most of the published literature related to the current design problem has focused on selecting design points $x_u \in [-1, 1]$, $u = 1, 2, \dots, N$, to estimate the slope of the quadratic model, $\alpha = \frac{d\eta}{dx} = \tau_1 - 2\tau_2 x$, or on estimating the turning point, with minimum variance. Ott

and Mendenhall (1972), Murty and Studden (1972) and Huda and Shafiq (1992) discussed optimal designs for estimating the slope of the quadratic model with maximum precision. Mandal (1978), Buonaccorsi and Iyer (1986), Chaloner (1989) and Fedorov and Müller (1997) discussed optimal designs for estimating the turning point with maximum precision. Melas et al. (2003) extended the criterion of Fedorov and Müller (1997) to derive truncated D-optimum designs for estimating the optimal point of quadratic regression models with minimum variance. Müller (1995) presented maximin efficient designs for estimating nonlinear functions in linear models with minimum variance. Atkinson et al. (2007) provided an excellent introduction to optimum experimental design.

Since the turning point and the corresponding optimal response are nonlinear functions of the parameters in the quadratic model, both variance and bias properties are important in their estimation. Therefore, deriving design criteria for variance and bias, as well as for the mean squared error, are important for the optimal design problem. Designs for estimating the optimum turning point with maximum accuracy, or equivalently with minimum bias, have not previously been addressed in the design literature. Similarly, designs for estimating the optimal response with minimum variance and bias have not received much attention, except for Müller (1995) who derived maximin efficient designs for estimating the optimal response with minimum variance. Atkinson et al. (1993) considered the estimation of the time until maximum concentration and the corresponding concentration from a compartmental model, with minimum variance.

In this paper we derive design criteria for selecting design points $x_u \in [-1, 1]$, $u = 1, 2, \dots, N$, to estimate γ and y_0 with minimum variance and bias. In Section 2 we present the optimum design criteria for estimating

nonlinear functions of the parameters in the quadratic model. In Sections 3 and 4 we derive optimum designs for estimating the turning point with minimum variance and bias, respectively. In Sections 5 and 6 we derive optimum designs for estimating the optimal response with minimum variance and bias, respectively. Results are illustrated graphically. Final conclusions are drawn in Section 7.

2. Optimum design criteria

An experimental design is considered to be a probability measure

$$\xi = \left\{ \begin{array}{cccc} x_1 & x_2 & \dots & x_N \\ p_1 & p_2 & \dots & p_N \end{array} \right\}$$

where the $x_u \in X$, $u = 1, 2, \dots, N$, are the design points with p_u , $u = 1, 2, \dots, N$, the associated design weights, i.e. the proportion of the total number of observations to be performed at the design points. The weights $p_u \in [0, 1]$ for all u and $\sum_{u=1}^N p_u = 1$. X denotes the design region of all the possible values of the x 's.

Define the quadratic model matrix $X = [f(\underline{x}_1)^T, f(\underline{x}_2)^T, \dots, f(\underline{x}_N)^T]^T$, where $f(\underline{x}_u)^T = (1, x_u, -x_u^2)$, $u = 1, 2, \dots, N$. Then, for the continuous design (defined below) the information matrix is approximated as

$$M(\xi) = \sum_{u=1}^N f(\underline{x}_u) f(\underline{x}_u)^T p_u. \quad (5)$$

The variance-covariance matrix of the least squares estimator $\hat{\underline{\tau}} = (\hat{\tau}_0, \hat{\tau}_1, \hat{\tau}_2)^T$ is $V(\hat{\underline{\tau}}) = \sigma^2 M^{-1}(\xi)$. Expressions $g_1(\hat{\underline{\tau}})$ and $g_2(\hat{\underline{\tau}})$ in (3) and (4) are nonlinear functions of the parameters in the quadratic model. Therefore, the optimal design criteria derived for estimating these nonlinear functions with minimum variance and bias, are functions of the information matrix $M(\xi)$, say

$\varphi(M(\xi))$. Using a continuous design measure eliminates the constraint that the number of trials at each design point must be an integer. Exact optimal designs may be found approximately by utilising continuous optimal designs (Atkinson et al., 2007).

The optimal design criteria derived in this paper are presented in Table 1. In (7) and (8) we define the vectors $\underline{c}_j^T = (c_{0j}(\underline{\tau}), c_{1j}(\tau), c_{2j}(\underline{\tau}))$, $j = 1, 2$, with elements:

$$c_{ij}(\underline{\tau}) = \frac{\partial g_j(\tau)}{\partial \tau_i}, \quad i = 0, 1, 2. \quad (6)$$

where $g_1(\tau)$ and $g_2(\tau)$, evaluated at $\hat{\tau} = \tau$, are defined in (3) and (4). The vectors $\underline{c}_1^T$ and $\underline{c}_2^T$ are specified in (14) and (25), respectively. Criteria (7) and (8) are known as c -optimality criteria for minimising the variance of a function of the parameters of the model, and will be referred to as $\varphi_V(\xi, \tau)$ -criteria. In accordance with the notation of Atkinson et al. (2007), designs which minimise the criterion $\varphi_V(\xi, \theta)$ for given $\underline{\tau}$ are termed C_τ -optimal. We assume that a best guess of $\underline{\tau}$, or equivalently γ , is available for constructing locally optimal designs. Criterion (7) is equivalent to the V_γ -optimal criterion discussed by Chaloner (1989) and Buonaccorsi and Iyer (1986). In (9) and (10), the elements of C_j , $j = 1, 2$, are the second-order derivatives of $g_j(\tau)$, $j = 1, 2$, with respect to the parameters in the quadratic model and are given in (22) and (30), respectively. Design criteria (9) and (10) are conveniently referred to as $\varphi_B(\xi, \tau)$ -criteria for minimising the squared bias of a function of the parameters of the model. Designs which minimise the criterion $\varphi_B(\xi, \tau)$ for given $\underline{\tau}$ are termed B_τ -optimal.

Table 1. Design criteria

Criterion for minimising the approximate:	Formulation	
variance of the turning point $\hat{\gamma}$	$\varphi_V(\xi, \tau) = \underline{c}_1^T M(\xi)^{-1} \underline{c}_1$	(7)
variance of the optimal response $\hat{y}_0$	$\varphi_V(\xi, \tau) = \underline{c}_2^T M(\xi)^{-1} \underline{c}_2$	(8)
squared bias of the turning point $\hat{\gamma}$	$\varphi_B(\xi, \tau) = \left(\frac{1}{2} \text{tr} \left(C_1^T M(\xi)^{-1} \right) \right)^2$	(9)
squared bias of the optimal response $\hat{y}_0$	$\varphi_B(\xi, \tau) = \left(\frac{1}{2} \text{tr} \left(C_2^T M(\xi)^{-1} \right) \right)^2$	(10)

For locally c -optimal designs there exists a powerful Equivalence Theorem which can be used to evaluate the global optimality or otherwise of a candidate design (Atkinson and Haines, 1996). Specifically, a design, ξ^* , is globally c -optimal if and only if the condition:

$$\left\{ f(x)^T M(\xi^*)^{-1} \underline{c} \right\}^2 - \underline{c}^T M(\xi^*)^{-1} \underline{c} \leq 0 \quad (11)$$

(the left-hand side of which is the directional derivative of the criterion function (Atkinson et al., 2007)) holds for all $x \in X$, with equality attained at the support points of ξ^* .

For Bayesian c -optimal designs the Equivalence Theorem states that a design, ξ^* , is globally c -optimal provided that the condition:

$$E \left\{ f(x)^T M(\xi^*)^{-1} \underline{c} \right\}^2 - E \left\{ \underline{c}^T M(\xi^*)^{-1} \underline{c} \right\} \leq 0 \quad (12)$$

is satisfied for all $x \in X$, where the expectation is taken over the prior distribution of the turning point $\gamma \in [-1, 1]$ (Atkinson and Haines, 1996). Using the criteria in Table 1, we derive symmetric and non-symmetric, non-singular designs, as well as Bayesian optimum designs. The Bayesian designs

will be referred to as AV - and AB -optimal designs for the variance and squared bias designs, respectively.

3. Variance optimum designs for the turning point

The estimation of the quadratic relationship in (2) requires at least three distinct levels of x . Consider the following design

$$\xi = \left\{ \begin{array}{ccc} -1 & s & 1 \\ p_1 & p_2 & p_3 \end{array} \right\} \quad (13)$$

where $s \in [-1, 1]$ and $p_u \in [0, 1]$, $u = 1, 2, 3$, the design weights, with $\sum_{u=1}^3 p_u = 1$. Many authors have considered design (13) with $s = 0$ in optimal design studies (Ott and Mendenhall, 1972; Murty and Studden, 1972; Mandal, 1978; Müller, 1995; Fedorov and Müller, 1997). Buonaccorsi and Iyer (1986) presented M -optimum designs, which minimises the maximum variance, for designs of the form (13) with s not necessarily equal to zero. Müller (1995) noted that Elfving's (1952) result provides that for estimating the turning point $\gamma = \frac{\tau_1}{2\tau_2}$ of the quadratic function, every locally optimal design has support with at least one point which is included in the set $\{-1, 0, 1\}$. Furthermore, the design with $s = 0$ and weight $1/3$ at each design point is the D - and G -optimal design for estimating the parameters in (2) with minimum variance (Atkinson et al., 2007). Therefore, it is good practice to consider designs with support on the points $\{-1, 0, 1\}$. In this article, we consider the derivation of optimum non-singular designs of the form (13) with $s = 0$ and $p_2 \neq 0$.

From criterion (7), the approximate variance of the turning point is $V(\hat{\gamma}) = \sigma^2 \underline{c}_1^T M(\xi)^{-1} \underline{c}_1$, where $\underline{c}_1^T = \left(0, \frac{1}{2\tau_2}, \frac{-\tau_1}{2\tau_2^2}\right) = \frac{1}{2\tau_2} (0, 1, -2\gamma)$ and

$M(\xi)$ is defined in (5). This simplifies to

$$V(\hat{\gamma}) = \frac{\sigma^2}{4\tau_2^2} \left(\frac{1}{\mu_2} + \frac{\left(2\gamma - \frac{1}{\mu_2}(\mu_3^* - \mu_1^*\mu_2^*)\right)^2}{(\mu_4^* - \mu_2^{*2}) - \frac{1}{\mu_2}(\mu_3^* - \mu_1^*\mu_2^*)^2} \right) \quad (14)$$

where $\mu_2 = \mu_2^* - \mu_1^{*2}$ (see Mandal, 1978), μ_r^* and μ_r are the r -th design moment of x about zero and about the mean, respectively. Note that the variance of the estimate of the slope of the quadratic model may, by standard likelihood theory, be approximated by $V(\hat{\alpha}) = \underline{c}_\alpha^T M^{-1}(\xi) \underline{c}_\alpha$, where $\hat{\alpha} = \hat{\tau}_1 - 2\hat{\tau}_2 x$ and $\underline{c}_\alpha = (0, 1, -2x)^T$. Therefore, locally c -optimal designs for minimising the variance of the optimum turning point for $\gamma \in [-1, 1]$ are equal to those for minimising the variance of the slope for $x \in [-1, 1]$ (Chaloner, 1989; Murty and Studden, 1972).

Optimum non-symmetric designs of the form (13), with $s = 0$, $p_1 \neq p_3$ and $p_2 \neq 0$ can be derived for $\gamma \in [-1, 1]$. Since $\mu_1^* = \mu_3^* = p_3 - p_1$ and $\mu_2 = p_1 + p_3 - (p_3 - p_1)^2$, the optimum design criterion for constructing the C_τ -optimal design is derived from (14) as:

$$\varphi_V(\xi, \tau) = \frac{1}{P_1} + \frac{\left(2\gamma - \frac{P_2}{P_1}\right)^2}{P_3 - \frac{P_2^2}{P_1}} \quad (15)$$

where

$$\begin{aligned} P_1 &= p_1 + p_3 - (p_3 - p_1)^2 \\ P_2 &= (p_3 - p_1)(1 - p_1 - p_3) \\ P_3 &= (p_1 + p_3)(1 - p_1 - p_3) \end{aligned}$$

From minimising (15), the following optimum weights are obtained (Coetzer, 2004):

$$\begin{aligned}
 & \text{(i) } p_1 = (2\gamma - 1) / (8\gamma), \quad p_2 = \frac{1}{2} \text{ and } p_3 = (2\gamma + 1) / (8\gamma) \text{ for } |\gamma| > \frac{1}{2} \\
 & \text{(ii) } p_1 = \frac{1}{2}, \quad p_2 = 2\gamma / (2\gamma - 1) \text{ and } p_3 = (1 + 2\gamma) / (2 - 4\gamma) \text{ for } -\frac{1}{2} < \gamma < 0 \\
 & \text{(iii) } p_1 = (1 - 2\gamma) / (2 + 4\gamma), \quad p_2 = 2\gamma / (2\gamma + 1) \text{ and } p_3 = \frac{1}{2} \text{ for } 0 < \gamma < \frac{1}{2}
 \end{aligned} \tag{16}$$

Note that, if $\gamma = 0$, we obtain a singular design since $p_2 = 0$. More observations are assigned to $x = 0$ than to $x = \pm 1$ for $|\gamma| > 0.5$, whereas the opposite holds for $|\gamma| < 0.25$.

The equivalence theorem in (11) can be used to evaluate the global optimality or otherwise of a candidate design. For non-symmetric C_τ -optimum designs of the form (13), with $s = 0$ and $p_1 \neq p_3$, $p_2 \neq 0$, with optimal weights (16), the derivative function on the left-hand side of (11) becomes:

$$\begin{aligned}
 & \left(\frac{-2\gamma \left((p_1 + p_3)^2 - (p_3 - p_1)^2 + P_2x - P_1x^2 \right) + P_3x - P_2x^2}{P_1P_3 - P_2^2} \right)^2 \\
 & - \left(\frac{1}{P_1} + \frac{(2\gamma - P_2/P_1)^2}{P_3 - P_2^2/P_1} \right)
 \end{aligned} \tag{17}$$

where P_i , $i = 1, 2, 3$, are given in (15). It is not difficult to show that condition (11) is satisfied for all $x \in [-1, 1]$, so the design is globally c -optimal.

Ott and Mendenhall (1972) considered minimising the variance of the slope $V(\hat{\tau}_1 - 2\hat{\tau}_2x)$ of the quadratic model. They evaluated practical design conditions which make the variance of the slope as small as possible at $x = 0$, and also minimise the rate of change of the variance with x there, or in an interval around $x = 0$, for example $|x| \leq \frac{1}{2}$. Therefore, as a special case, if we invoke the same arguments used by Ott and Mendenhall (1972); i.e.

minimising $V(\hat{\gamma})$ as well as the rate of change of the variance at $\gamma = 0$, or in an interval around $\gamma = 0$, we obtain the optimum symmetric design with $s = 0$ and $p_1 = p_3 = P/2$, where $P \in [0, 1]$ is the optimal design weight (Coetzer, 2004). For this design, $P = \mu_2^*$ and the optimum design criterion for constructing the C_τ -optimal design is derived from (14) as:

$$\varphi_V(\xi, \tau) = \frac{1}{\mu_2^*} + \frac{4\gamma^2}{\mu_2^*(1 - \mu_2^*)} \quad (18)$$

The optimal solution for (18) is

$$\mu_2^* = 1 + 4\gamma^2 - 2\gamma\sqrt{1 + 4\gamma^2} \text{ for } \gamma \neq 0 \quad (19)$$

Expression (19) provides the optimal design weights, since $p_1 = p_3 = P/2 = \mu_2^*/2$. Note that, if $\gamma = 0$, $P = 1$, which yields a singular design with equal weights $\frac{1}{2}$ at $x = \pm 1$. For $0 < |\gamma| < 0.2$, more observations are assigned to $x = \pm 1$ than to $x = 0$, whereas the opposite holds otherwise. For c -optimal designs, the efficiency of a design ξ can be defined as

$$Eff(\xi) = \frac{\underline{c}^T M(\xi^*)^{-1} \underline{c}}{\underline{c}^T M(\xi)^{-1} \underline{c}} 100\% \quad (20)$$

where ξ^* is the optimal design (Atkinson et al., 1993). The efficiency measure is directly related to the number of observations; an efficiency of 50% indicates that double the sample size is required to make the design as powerful as the optimal design for estimating the property of interest. The efficiency for the symmetric C_τ -optimal design (19) relative to the non-symmetric C_τ -optimal design (16) may be shown never to fall below 72%, which indicates that the symmetrical C_τ -optimal design is very efficient for estimating the turning point for $\gamma \in [-1, 1]$, especially for $|\gamma| < 0.25$ where it is greater than 90%.

The symmetric optimal design, with weight $P/2$ at $x = \pm 1$, corresponds to the design with a uniform prior on $\gamma \in [-1, 1]$ (Chaloner, 1989). A Bayesian AV-optimal design can be derived from criterion (18). $V(\gamma) = E(\gamma^2) = 1/3$ for a uniform prior on $\gamma \in [-1, 1]$. Minimising the posterior expectation $E(V(\hat{\gamma}))$ with respect to P we obtain optimal $P = 0.5695$. Therefore, the AV-optimal design puts weight 0.28475 at ± 1 and weight 0.4305 at 0. Mandal (1978) also obtained these results using different arguments. The equivalence theorem in (12) can be used to evaluate the global optimality or otherwise of the Bayesian AV-optimal design. For Bayesian AV-optimal designs for the turning point, the left-hand side of (12) becomes:

$$\frac{E(\gamma^2)(P^2 - Px^2)^2 + x^2P^2(1-P)^2}{P^4(1-P)^2} - \frac{1-P+4E(\gamma^2)}{P(1-P)} \quad (21)$$

where the expectation is taken over a prior distribution for $\gamma \in [-1, 1]$. It is straight-forward to show that expression (21) confirms the global optimality of the AV-optimal design for estimating the turning point for a uniform prior on $\gamma \in [-1, 1]$. If, however, $E(\gamma) = 0$ and $V(\gamma) = 1/12$ then the Bayesian criterion puts mass 1/3 at each of the design points $(-1, 0, 1)$, which is also the D-optimal design for estimating the parameters of the quadratic model in (2). The uniform prior on $[-\frac{1}{2}, \frac{1}{2}]$ is such a prior (Chaloner, 1989).

4. Bias optimum designs for the turning point

$\text{Bias}(\hat{\gamma}) = E(\hat{\gamma}) - \gamma$ indicates the accuracy of $\hat{\gamma}$. From criterion (9) the approximate bias of the turning point is $\text{Bias}(\hat{\gamma}) = \frac{\sigma^2}{2} \text{tr}(C_1 M(\xi)^{-1})$, where

$$C_1 = \frac{1}{\tau_2^2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 2\gamma \end{pmatrix}. \text{ This simplifies to}$$

$$\text{Bias}(\hat{\gamma}) = \frac{\sigma^2}{\tau_2^2} \left(\frac{\gamma - \frac{1}{2\mu_2} (\mu_3^* - \mu_1^* \mu_2^*)}{(\mu_4^* - \mu_2^{*2}) - \frac{1}{\mu_2} (\mu_3^* - \mu_1^* \mu_2^*)^2} \right) \quad (22)$$

The derivation of expression (22) is given in Appendix A. Since $\text{Bias}(\hat{\gamma})$ is signed, we minimise the squared bias criterion, $\varphi_B(\xi, \tau) = \text{Bias}(\hat{\gamma})^2$.

It may be argued that the variance dominates the bias and that the latter is not important for statistical estimation. Therefore, we first motivate the relevance of the squared bias as a design criterion. Figure 1 depicts the optimum variance and squared bias criteria for the symmetric and non-symmetric, non-singular designs, for selected values of $\gamma \in [-1, 1]$. The optimal weights for the squared bias criterion will be presented later in this section. For the symmetric designs, it can be observed from Figure 1 that the squared bias criterion, $\varphi_B(\xi, \tau)$, is very similar in magnitude compared to the variance criterion $\varphi_V(\xi, \tau)$, although $\varphi_B(\xi, \tau)$ tends to zero as γ approaches zero from below and above. For the non-symmetric designs, the magnitude of the squared bias criterion, $\varphi_B(\xi, \tau)$, and that of the variance criterion $\varphi_V(\xi, \tau)$ are also comparable, except for $|\gamma| < 0.4$ where $\varphi_B(\xi, \tau)$ equals zero. The squared bias and variance criteria are surprisingly close as γ approaches ± 1 . In view of the fact that the mean squared error (MSE) equals the sum of these two criteria, these results clearly illustrate that the bias is as important as the variance in estimating the optimal turning point, and hence the relevance of studying optimum squared bias designs.

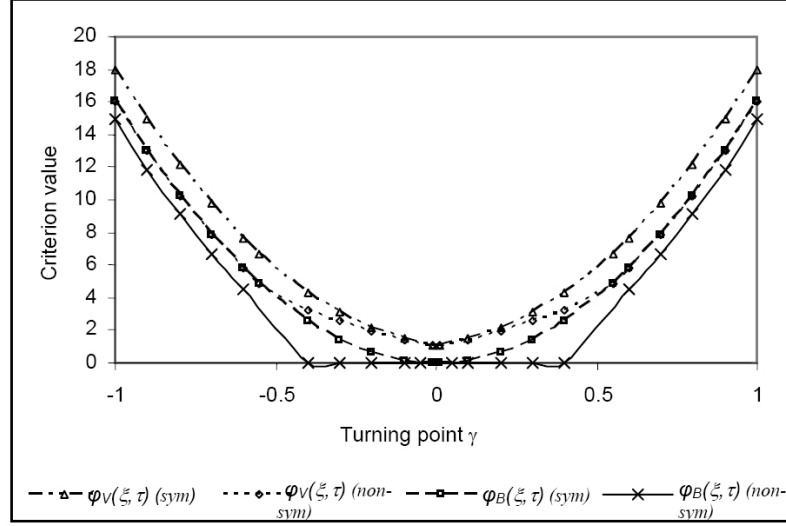


Figure 1. Comparisons of the magnitude of the variance and squared bias criteria for the symmetric and non-symmetric, non-singular designs

Optimum non-symmetric squared bias designs are derived for designs of the form (13) with $s = 0$, $p_i \neq 0$, $i = 1, 2, 3$ for $\gamma \in [-1, 1]$. From expression (22) the optimum non-symmetric B_τ -optimal design may be expressed as:

$$\varphi_B(\xi, \tau) = \frac{\left(\gamma - \frac{P_2}{2P_1}\right)^2}{\left(P_3 - \frac{P_2^2}{P_1}\right)^2} \quad (23)$$

where P_i , $i = 1, 2, 3$, are given in (15). The optimal weights, obtained by optimising criterion (23) numerically, are plotted in Figure 2 for different values of $\gamma \in [-1, 1]$. Note that, as with the non-symmetric variance criterion, the squared bias is not defined for $\gamma = 0$, which yields a singular design.

Whereas analytical expressions (16) could be obtained for the optimal p_1 , p_2 and p_3 for minimising the non-symmetric variance criterion (15), optimal weights for the non-symmetric bias criterion (23) could only be obtained numerically. From Figure 2, we observe that almost all the weight is placed on the centre point for $|\gamma| \leq \frac{1}{2}$. However, for $|\gamma| > \frac{1}{2}$, $p_2 \approx 0.5$ which is equal to the corresponding design weight at the centre point for the symmetrical squared bias criterion, discussed below.

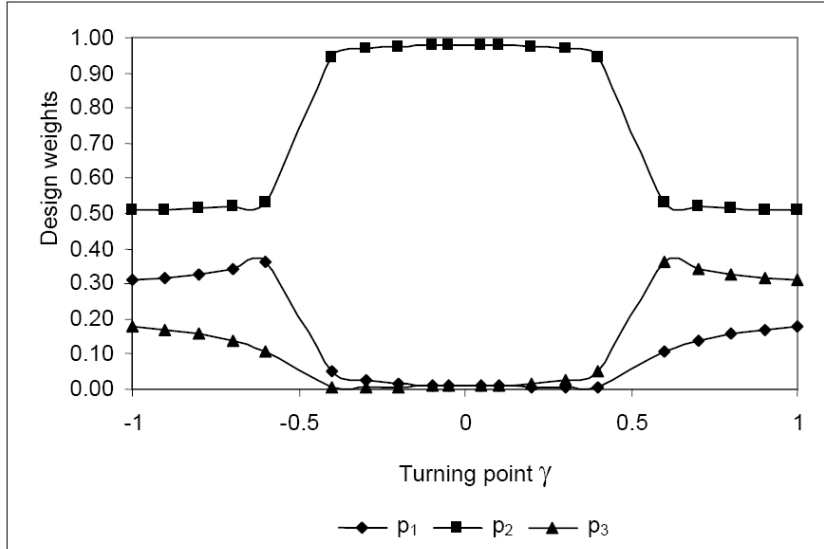


Figure 2. Optimum design weights plotted against the true turning point for the non-symmetric B_τ -optimal design (23). (p_1 = weight at $x = -1$, p_2 = weight at $x = 0$, p_3 = weight at $x = +1$).

The dashed line in Figure 3 depicts the efficiency of the C_τ (variance)-optimal design with weights (16) relative to the B_τ (bias)-optimal design using the efficiency measure (20). The C_τ -optimal design is very efficient in minimising the squared bias for $|\gamma| > \frac{1}{2}$. Therefore, the C_τ -optimal design, which is the global optimal design for estimating the turning point with minimum variance, is also very efficient for estimating it with minimum squared bias when $|\gamma| > \frac{1}{2}$. However, for $|\gamma| < \frac{1}{2}$, the efficiency of the C_τ -optimal design decreases rapidly to zero.

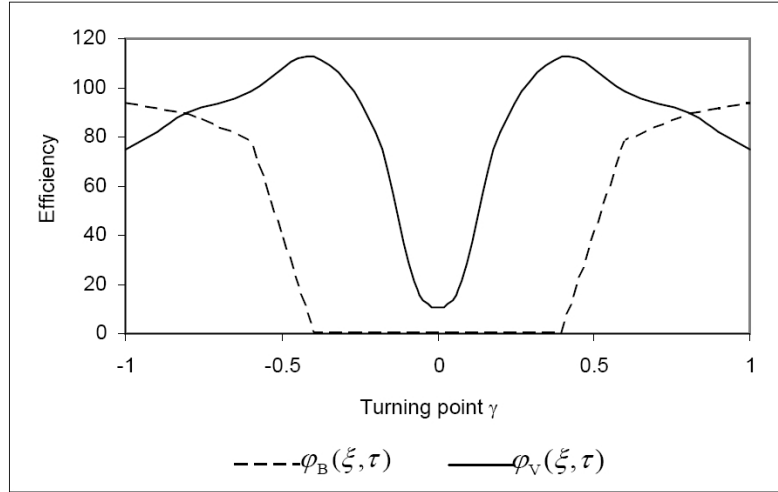


Figure 3. Efficiency plot for the C_τ -optimal design (16) for minimising the $\varphi_B(\xi, \tau)$ -criterion (23) for the turning point, and the $\varphi_V(\xi, \tau)$ -criterion for the optimal response (26).

For symmetrical designs of the form (13), with $s = 0$ and $p_1 = p_3 = P/2$, the optimum design criterion for constructing the B_τ -optimal design, is derived

from (23) as:

$$\varphi_B(\xi, \tau) = \frac{\gamma^2}{\mu_2^{*2} (1 - \mu_2^*)^2} \quad (24)$$

Criterion (24) yields optimum weight $P = \frac{1}{2}$, since $P = \mu_2^*$. This symmetric, bias-optimal design is completely defined by $P = \frac{1}{2}$ and can therefore be referred to as a global optimal design. As discussed earlier, the symmetric design corresponds to the design with a uniform prior on $\gamma \in [-1, 1]$. The AB -optimal design, i.e. the Bayesian squared bias design, which minimises the posterior expectation of $\text{Bias}(\hat{\gamma})^2$, also yields the global optimal $P = \frac{1}{2}$ for a uniform prior on $\gamma \in [-1, 1]$.

The $\varphi_B(\xi, \tau)$ -criterion (24) is plotted against γ in Figure 4 for the B_τ -optimal design with weight $P = \frac{1}{2}$. The optimal weights in (19) for the symmetric C_τ -optimal design is also plotted in Figure 4 together with the AV -optimal design, with $P = 0.5695$ for a uniform prior on $\gamma \in [-1, 1]$, and the variance D-optimal design with $P = 2/3$, discussed earlier. All the designs yield a squared bias of zero at $\gamma = 0$ (except for the symmetric C_τ -optimal design which is not defined there). As expected, the B_τ -optimal design yields the smallest $\text{Bias}(\hat{\gamma})^2$ throughout the design range $\gamma \in [-1, 1]$. This design and the AV -optimal design with $P = 0.5695$ yield similar $\text{Bias}(\hat{\gamma})$ values for $|\gamma| < \frac{1}{2}$, whereas it yields similar values as the symmetric C_τ -optimal design for $|\gamma| > \frac{1}{2}$. The D-optimal design with $P = 2/3$ does not perform well in terms of the squared bias criterion for $\gamma \neq 0$.

From the Bayesian AV and AB optimum designs, we derive the AMSE design, i.e. the optimal Bayesian mean square error (MSE) design. The AMSE design minimises the posterior expectation of the MSE, which, under a uniform

prior on $\gamma \in [-1, 1]$ is given by:

$$E(MSE(\hat{\gamma})) = E(V(\hat{\gamma})) + E(\text{Bias}(\hat{\gamma})^2) = \frac{7/3 - P}{4P(1-P)} + \frac{1}{3P^2(1-P)^2}.$$

Minimising $E(MSE(\hat{\gamma}))$ with respect to P yields optimal $P = 0.51$. Note that the optimal weight for the AV -design is $P = 0.5695$, which indicates that the variance does not dominate the MSE because the optimal weight for the AB -design is $P = 0.5$, which is very similar to that of the $AMSE$ design.

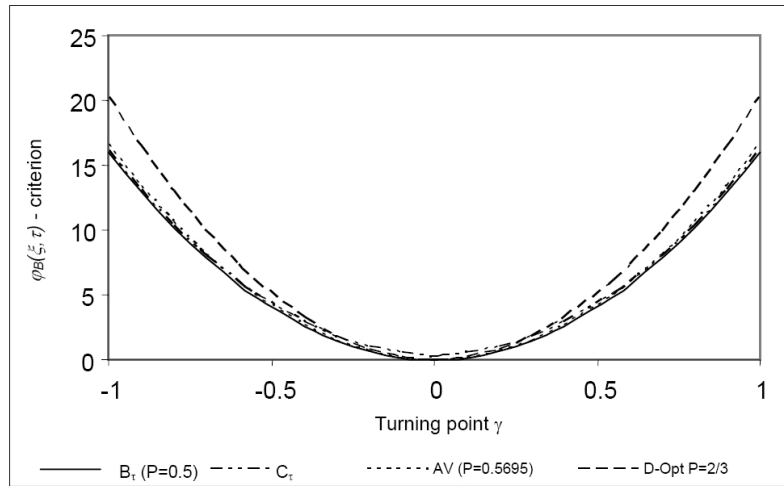


Figure 4. The symmetric $\varphi_B(\xi, \tau)$ -criterion (24) for the turning point plotted against the true turning point for five different symmetrical designs: The B_τ -optimal design (24); C_τ -optimal design (19); AV -optimal design with $P = 0.5695$ using a uniform prior on $\gamma \in [-1, 1]$; Variance D -optimal design for parameter estimation with $P = 2/3$.

5. Variance optimum designs for the optimal response

From criterion (8) the approximate variance of the optimal response is $V(\hat{y}_0) = \sigma^2 \underline{c}_2^T M(\xi)^{-1} \underline{c}_2$, where $\underline{c}_2^T = (1, \gamma, -\gamma^2)$. This becomes:

$$V(\hat{y}_0) = \sigma^2 \frac{\{(\mu_2^* \mu_4^* - \mu_3^{*2}) - 2\gamma^2 (\mu_2^{*2} - \mu_1^* \mu_3^*) + 2\gamma (\mu_2^* \mu_3^* - \mu_1^* \mu_4^*)\}}{\{(\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_2^* \mu_1^*)^2\}} + \sigma^2 \left\{ \gamma^2 \left[\frac{1}{\mu_2} + \frac{\left(\gamma - \frac{1}{\mu_2} (\mu_3^* - \mu_2^* \mu_1^*) \right)^2}{\left\{ (\mu_4^* - \mu_2^{*2}) - \frac{1}{\mu_2} (\mu_3^* - \mu_2^* \mu_1^*)^2 \right\}} \right] \right\} \quad (25)$$

The derivation of expression (25) is given in Appendix B.

Optimum non-symmetric designs can be derived for designs of the form (13) with $s = 0$, $p_1 \neq p_3$ and $\gamma \in [-1, 1]$. As previously, we consider all three distinct design points with $p_2 \neq 0$. The optimum design criterion for constructing the C_τ -optimal design is derived from (25) as:

$$\varphi_V(\xi, \tau) = \frac{(1 - 2\gamma^2) ((p_1 + p_3))^2 - (p_3 - p_1)^2}{P_1 \left(P_3 - \frac{P_2^2}{P_1} \right)} + \gamma^2 \left(\frac{1}{P_1} + \frac{\left(\gamma - \frac{P_2}{P_1} \right)^2}{P_3 - \frac{P_2^2}{P_1}} \right) \quad (26)$$

where P_i , $i = 1, 2, 3$, are given in (15). The optimum weights, obtained from optimising the $\varphi_V(\xi, \tau)$ -criterion (26) for different values of $\gamma \in [-1, 1]$, are plotted against γ in Figure 5. Note that the optimum solutions could only be obtained numerically. More observations are assigned to the centre point as γ approaches zero.

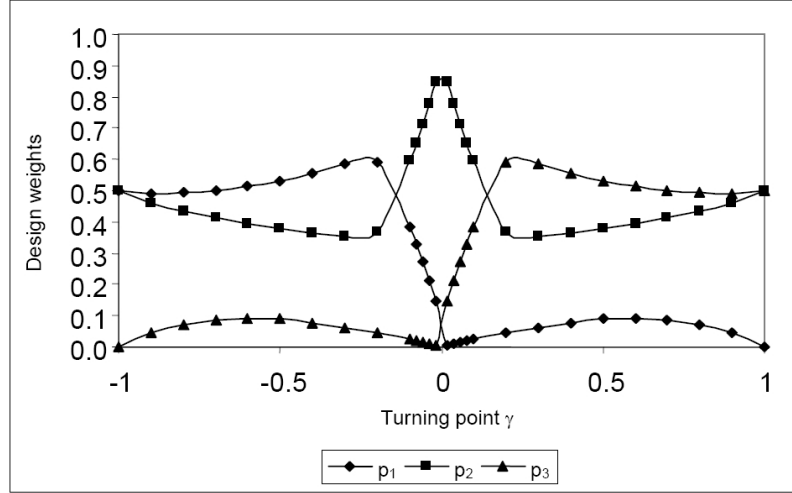


Figure 5. Optimum design weights plotted against the true turning point for minimising the variance of the optimal response (non-symmetric $\varphi_V(\xi, \tau)$ -optimal design (26) with $p_1 =$ weight at -1 , $p_2 =$ weight at 0 , $p_3 =$ weight at $+1$).

Figure 3 depicts the efficiency of the C_τ -optimal design for the turning point with weights given in (16), for minimising the $\varphi_V(\xi, \tau)$ variance criterion for the optimal response, using efficiency measure (20). The C_τ -optimal design for the turning point is very efficient in estimating the optimal response with minimum variance for $|\gamma| > 0.25$, but is very inefficient when $|\gamma| < 0.25$. Note that for $|\gamma|$ around 0.4, the efficiency of the C_τ -optimal design appears to be greater than 100%. However, this might be an artefact of the optimisation algorithm which, in some cases, could not guarantee a true minimum. However, Figure 5 presents the optimal weights for the C_τ -optimal design for the optimal response, which should be employed in practice, for non-symmetric designs of the form (13).

For symmetrical designs of the form (13), with $s = 0$ and $p_1 = p_3 = P/2$, the optimum design criterion for constructing the symmetric C_τ -optimal design is derived from (26) as:

$$\varphi_V(\xi, \tau) = \frac{\gamma^2}{\mu_2^*} + \frac{1 - 2\gamma^2}{1 - \mu_2^*} + \frac{\gamma^4}{\mu_2^*(1 - \mu_2^*)} \quad (27)$$

The optimal solution is:

$$\mu_2^* = \frac{\gamma^2(\gamma^2 + 1)}{1 - 3\gamma^2} \left(-1 - \frac{\gamma^2 - 1}{\gamma(\gamma^2 + 1)^{1/2}} \right) \text{ for } \gamma \neq 0 \quad (28)$$

Expression (28) provides the optimal design weights. Note that, if $\gamma = 0$, $P = \mu_2^* = 0$, which yields a singular design since all the weight is assigned to the centre point. Also, if $\gamma = 1$, $P = \mu_2^* = 1$, which again yields a singular design with equal weight $\frac{1}{2}$ assigned to $x = \pm 1$. Thus we do not define the optimal design for $\gamma = 0$ or 1.

The weights for the symmetric C_τ -optimal design for the optimal response (denoted $C_\tau(y)$) are plotted against γ in Figure 6 together with those for the symmetric optimal design for the turning point (denoted $C_\tau(\gamma)$) given in (19). For the C_τ -optimal design for the optimal response, the optimal weight $P/2$ at $x = \pm 1$ decreases approximately linearly to zero, and almost all the weight is assigned to the centre point, as γ approaches 0. In contrast, for the C_τ -optimal design for the turning point, the proportion of observations assigned to the centre point tends to zero when γ is close to 0 and almost all the weight is assigned to $x = \pm 1$. In general, more observations must be assigned to the centre point than to the end points when designing for the optimal response, in contrast to when designing for the turning point.

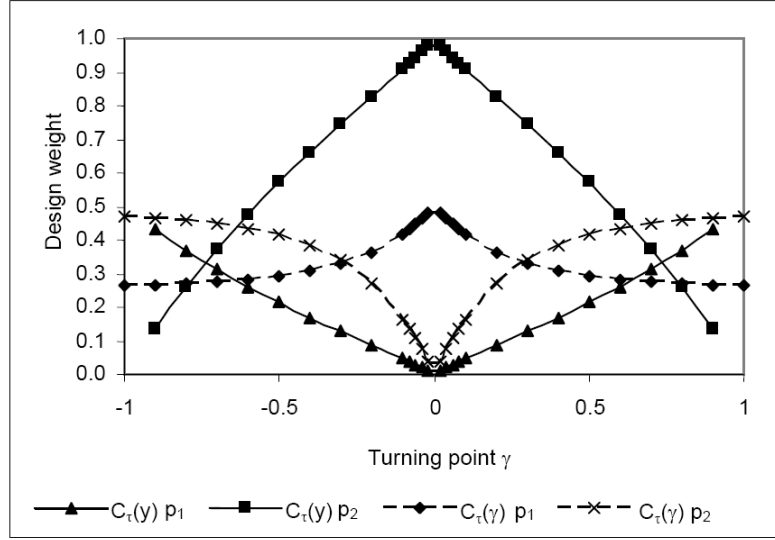


Figure 6. Optimum design weights plotted against the true turning point for minimising the variance of the optimal response (symmetric C_τ -optimal design, $C_\tau(y)$ (28)) together with the optimum weights for minimising the variance of the turning point (symmetric C_τ -optimal design, $C_\tau(\gamma)$ (19)).
 $(p_1 = p_3 = \text{weight at } \pm 1, p_2 = \text{weight at } 0)$.

The symmetric C_τ -optimal design under study corresponds to the design with a uniform prior on $\gamma \in [-1, 1]$. Therefore, a Bayesian-type AV -optimal design can also be derived from criterion (27) for estimating the optimal response. For a uniform prior on $\gamma \in [-1, 1]$, $V(\gamma) = E(\gamma^2) = 1/3$ and $E(\gamma^4) = 1/5$. Thus, the posterior expectation of $V(\hat{y}_0)$ can be written from (27) as:

$$E(V(\hat{y}_0)) = \frac{8}{15P(1-P)} \quad (29)$$

since $P = \mu_2^*$. Minimising $E(V(\hat{y}_0))$ with respect to P we obtain optimal $P = 0.5$. Therefore, the AV -optimal design, for minimising the posterior expectation of $V(\hat{y}_0)$ under a uniform prior on γ , puts weight $\frac{1}{4}$ at ± 1 and

weight 0.5 at 0. So, the AV -optimal design for minimising the variance of the optimal response is the same as the AB -optimal design for minimising the posterior expectation of the squared bias of the estimated turning point, $E(\text{Bias}(\hat{\gamma})^2)$, under a uniform prior on $\gamma \in [-1, 1]$.

6. Bias optimum designs for the optimal response

The accuracy of the estimate $\hat{y}_0$ is expressed by $\text{Bias}(\hat{y}_0) = E(\hat{y}_0) - \hat{y}_0$.

From criterion (10) the approximate bias of the optimal response is $\text{Bias}(\hat{y}_0) = \frac{\sigma^2}{2} \text{tr} \left(C_2 M(\xi)^{-1} \right)$, where $C_2 = \frac{1}{\tau_2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -\gamma \\ 0 & -\gamma & 4\gamma^2 \end{pmatrix}$. This simplifies to

$$\text{Bias}(\hat{y}_0) = \tau_2 V(\hat{\gamma}), \text{ where } \tau_2 \text{ is defined in (2).} \quad (30)$$

The proof of (30) is given in Appendix C. The bias of the estimated optimal response is therefore proportional to the variance of the estimated turning point. Thus the optimal designs for minimising these two criteria are the same, and are given in (16). Furthermore, for non-symmetric designs, the globally C_τ -optimal design in (16) will also be globally optimal for estimating the optimal response with minimum bias. Note that, since $\tau_2 > 0$ from (2), $\text{Bias}(\hat{y}_0) \geq 0$ for the maximum response. Therefore, $\hat{y}_0$ tends to over-estimate the true optimal response. From a practical perspective, this means that the estimated optimal response might not be achievable, and that a more conservative target should be set. Furthermore, the equivalence of these two criteria shows that a design aimed at the precise estimation of the turning point will also minimise the bias of the estimated optimal response.

An AB -optimal design can be derived for the optimal response for a uniform prior on $\gamma \in [-1, 1]$. The posterior expectation of the squared bias

under this prior can be derived from that of $V(\hat{\gamma}^2)$ as:

$$E(\text{Bias}(\hat{y}_0)^2) = E(V(\hat{\gamma})^2) = \frac{P^2 - 14P/3 + 103/15}{P^2(1-P)^2}$$

Minimising this yields optimal $P = 0.548$. In contrast, minimising the posterior expectation of the bias yields optimal $P = 0.5695$, which is equal to the AV -optimal design for the turning point. The above expression for $E(\text{Bias}(\hat{y}_0)^2)$ may be used for evaluating the posterior expectation of $\text{MSE}(\hat{y}_0)$:

$$\begin{aligned} E(\text{MSE}(\hat{y}_0)) &= E(V(\hat{y}_0)) + \tau_2^2 E(V(\hat{\gamma})^2) \\ &= \frac{8}{15P(1-P)} + \frac{P^2 - 14P/3 + 103/15}{16\tau_2^2 P^2(1-P)^2} \end{aligned}$$

The AMSE design, i.e. the Bayesian MSE design, is obtained by minimising this posterior expectation. $E(\text{MSE}(\hat{y}_0))$ depends on the parameter τ_2 , the coefficient of x^2 in the response surface (2). For example, minimising $E(\text{MSE}(\hat{y}_0))$ with respect to P yields optimal $P = 0.5391$ for $\tau_2 = 1$, for a uniform prior on $\gamma \in [-1, 1]$.

7. Conclusion

In this paper various experimental designs were derived for estimating the optimal turning point, and the corresponding optimal response, from a one-factor quadratic model, with maximum precision and accuracy. Criteria were derived for the variance and the squared bias of the estimates of both the turning point and the optimal response, in terms of the design moments of the information matrix. Utilising continuous optimal design theory, locally c -optimum designs were specified for both symmetrical and non-symmetrical, non-singular designs. Bayesian optimum designs were also derived assuming

a uniform prior over the design range for the optimal turning point. While these designs may be more practical to deploy than the locally c -optimum designs, because they do not depend on point estimates of the parameters in the response surface model, they do depend on the uniform prior assumption.

Designs for the squared bias of the optimal turning point, as well as for the variance and the bias of the optimal response, have not received much attention in the literature. Although the magnitude of the squared bias will typically be smaller than the variance of the estimate, design criteria for the squared bias and the variance provide different options for constructing optimal designs.

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Appendix A. Derivation of the approximate bias of the turning point

The r -th design moment of x about zero and about the mean, respectively, are defined as $\mu_r^* = \sum_{u=1}^N x_u^r p_u$ and $\mu_r = \sum_{u=1}^N (x_u - \mu_1^*)^r p_u$. The approximate Bias($\hat{\gamma}$) in (22) is derived as follows:

$$\begin{aligned}
 \text{Bias}(\hat{\gamma}) &= E(\hat{\gamma}) - \gamma \\
 &= \frac{\sigma^2 \tau_1}{2\tau_2^3} \mu_2 \left((\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_1^* \mu_2^*)^2 \right)^{-1} \\
 &\quad - \frac{\sigma^2}{2\tau_2^2} \left((\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_1^* \mu_2^*)^2 \right)^{-1} (\mu_3^* - \mu_2^* \mu_1^*) \\
 &= \frac{\sigma^2}{\tau_2^2} \left((\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_1^* \mu_2^*)^2 \right)^{-1} \\
 &\quad \cdot \left(\frac{\tau_1}{2\tau_2} \mu_2 - \frac{1}{2} (\mu_3^* - \mu_3^* - \mu_1^* \mu_2^*) \right) \\
 &= \frac{\sigma}{\tau_2^2} \left(\frac{\mu_2 \gamma - \frac{1}{2} (\mu_3^* - \mu_1^* \mu_2^*)}{(\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_1^* \mu_2^*)^2} \right) \\
 &= \frac{\sigma}{\tau_2^2} \left(\frac{\gamma - \frac{1}{2\mu_2} (\mu_3^* - \mu_1^* \mu_2^*)}{(\mu_4^* - \mu_2^{*2}) - \frac{1}{\mu_2} (\mu_3^* - \mu_1^* \mu_2^*)^2} \right)
 \end{aligned}$$

Appendix B. Derivation of the approximate variance of the optimal response

The approximate variance (25) is derived as follows

$$V(\hat{y}_0) = \sigma^2 \left((\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_2^* \mu_1^*)^2 \right)^{-1}$$

$$\begin{aligned}
& \begin{pmatrix} -\frac{\tau_1^2}{4\tau_2^2} & \frac{\tau_1}{2\tau_2} \end{pmatrix} \begin{pmatrix} \mu_2^*\mu_4^* - \mu_3^{*2} & \mu_2^{*2} - \mu_1^*\mu_3^* & \mu_2^*\mu_3^* - \mu_1^*\mu_4^* \\ \mu_2^{*2} - \mu_1^*\mu_3^* & \mu_2 & \mu_3^* - \mu_2^*\mu_1^* \\ \mu_2^*\mu_3^* - \mu_1^*\mu_4^* & \mu_3^* - \mu_2^*\mu_1^* & \mu_4^* - \mu_2^{*2} \end{pmatrix} \begin{pmatrix} 1 \\ -\frac{\tau_1^2}{4\tau_2^2} \\ \frac{\tau_1}{2\tau_2} \end{pmatrix} \\
&= \sigma^2 \left((\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_2^*\mu_1^*)^2 \right)^{-1} \begin{pmatrix} \mu_2^*\mu_4^* - \mu_3^{*2} - \frac{\tau_1^2}{2\tau_2^2} (\mu_2^{*2} - \mu_1^*\mu_3^*) \\ + \frac{\tau_1}{\tau_2} (\mu_2^*\mu_3^* - \mu_1^*\mu_4^*) + \frac{\tau_1^4}{16\tau_2^4} \mu_2 \\ - \frac{\tau_1^3}{4\tau_2^3} (\mu_3^* - \mu_2^*\mu_1^*) \\ + \frac{\tau_1^2}{4\tau_2^2} (\mu_4^* - \mu_2^{*2}) \end{pmatrix} \\
&= \sigma^2 \frac{\left\{ \begin{aligned} & \mu_2^*\mu_4^* - \mu_3^{*2} - 2\gamma^2 (\mu_2^{*2} - \mu_1^*\mu_3^*) + 2\gamma (\mu_2^*\mu_3^* - \mu_1^*\mu_4^*) + \gamma^4 \mu_2 - 2\gamma^3 \\ & (\mu_3^* - \mu_2^*\mu_1^*) + \gamma^2 (\mu_4^* - \mu_2^{*2}) \end{aligned} \right\}}{\left\{ (\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_2^*\mu_1^*)^2 \right\}}
\end{aligned}$$

Simplifying further we obtain

$$\begin{aligned}
V(\hat{y}_0) &= \sigma^2 \frac{\left\{ (\mu_2^*\mu_4^* - \mu_3^{*2}) - 2\gamma^2 (\mu_2^{*2} - \mu_1^*\mu_3^*) + 2\gamma (\mu_2^*\mu_3^* - \mu_1^*\mu_4^*) \right\}}{\left\{ (\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_2^*\mu_1^*)^2 \right\}} \\
&+ \sigma^2 \frac{\left\{ \gamma^2 [\gamma^2 \mu_2 - 2\gamma (\mu_3^* - \mu_2^*\mu_1^*) + (\mu_4^* - \mu_2^{*2})] \right\}}{\left\{ (\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_2^*\mu_1^*)^2 \right\}} \\
&= \sigma^2 \frac{\left\{ (\mu_2^*\mu_4^* - \mu_3^{*2}) - 2\gamma^2 (\mu_2^{*2} - \mu_1^*\mu_3^*) + 2\gamma (\mu_2^*\mu_3^* - \mu_1^*\mu_4^*) \right\}}{\left\{ (\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_2^*\mu_1^*)^2 \right\}} \\
&+ \sigma^2 \left\{ \gamma^2 \left[\frac{1}{\mu_2} + \frac{\left(\gamma - \frac{1}{\mu_2} (\mu_3^* - \mu_2^*\mu_1^*) \right)^2}{\left\{ (\mu_4^* - \mu_2^{*2}) - \frac{1}{\mu_1} (\mu_3^* - \mu_2^*\mu_1^*)^2 \right\}} \right] \right\}
\end{aligned}$$

Appendix C. Derivation of the approximate bias of the optimal response

The approximate Bias($\hat{y}_0$) in (30) is derived as follows

$$\begin{aligned}
\text{Bias}(\hat{y}_0) &= E(\hat{y}_0) - y_0 \\
&= \sigma^2 \frac{\left\{ \frac{\tau_1^2}{4\tau_2^3} \mu_2 + \frac{1}{4\tau_2} (\mu_4^* - \mu_2^{*2}) - \frac{\tau_1}{2\tau_2^2} (\mu_3^* - \mu_2^* \mu_1^*) \right\}}{\left((\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_2^* \mu_1^*)^2 \right)} \\
&= \frac{\sigma^2}{4\tau_2} \frac{\left\{ \frac{\tau_1^2}{\tau_2^2} \mu_2 + (\mu_4^* - \mu_2^{*2}) - \frac{2\tau_1}{\tau_2} (\mu_3^* - \mu_2^* \mu_1^*) \right\}}{\left((\mu_4^* - \mu_2^{*2}) \mu_2 - (\mu_3^* - \mu_2^* \mu_1^*)^2 \right)} \\
&= \frac{\sigma^2}{4\tau_2} \left\{ \frac{(\mu_4^* - \mu_2^{*2}) - \frac{1}{\mu_2} (\mu_3^* - \mu_2^* \mu_1^*)^2}{\mu_2 \left((\mu_4^* - \mu_2^{*2}) - \frac{1}{\mu_2} (\mu_3^* - \mu_2^* \mu_1^*)^2 \right)} \right. \\
&\quad \left. + \frac{4\gamma^2 - \frac{4\gamma}{\mu_2} (\mu_3^* - \mu_2^* \mu_1^*) + \frac{1}{\mu_2^2} (\mu_3^* - \mu_2^* \mu_1^*)^2}{(\mu_4^* - \mu_2^{*2}) - \frac{1}{\mu_2} (\mu_3^* - \mu_2^* \mu_1^*)^2} \right\} \\
&= \frac{\sigma^2}{4\tau_2} \left\{ \frac{1}{\mu_2} + \frac{\left(2\gamma - \frac{1}{\mu_2} (\mu_3^* - \mu_2^* \mu_1^*) \right)^2}{(\mu_4^* - \mu_2^{*2}) - \frac{1}{\mu_2} (\mu_3^* - \mu_2^* \mu_1^*)^2} \right\} \\
&= \tau_2 V(\hat{\gamma})
\end{aligned}$$